Notes on Model Evaluation Metrics:

– **Accuracy** is the proportion of all classifications that were correct, whether positive or negative.

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

– A perfect model would have zero false positives and zero false negatives, and therefore an accuracy of 1.0, or 100%.

– For heavily imbalanced datasets, where one class appears very rarely, say 1% of the time, a model that predicts negative 100% of the time would score 99% on accuracy, despite being useless.

– The **true positive rate (TPR)**, or the proportion of all actual positives that were classified correctly as positives, is also known as **recall**.

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

– In an imbalanced dataset where the number of actual positives is very low, recall is a more meaningful metric than accuracy because it measures the ability of the model to correctly identify all positive instances.

– The **false positive rate (FPR)** is the proportion of all actual negatives that were classified *incorrectly* as positives, also known as the **probability of false alarm.**

$$\text{FPR} = \frac{\text{incorrectly classified actual negatives}}{\text{all actual negatives}} = \frac{FP}{FP + TN}$$

– **Precision** is the proportion of all the model's positive classifications that are actually positive.

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

– a more stable metric for evaluating the effects of false positives.

– In an imbalanced dataset where the number of actual positives is very, very low, say 1-2 examples in total, precision is less meaningful and less useful as a metric.

– Precision improves as false positives decrease, while recall improves when false negatives decrease.

Metric	Guidance
Accuracy	Use as a rough indicator of model training progress/convergence for balanced datasets. For model performance, use only in combination with other metrics. Avoid for imbalanced datasets. Consider using another metric.
Recall (True positive rate)	Use when false negatives are more expensive than false positives.
False positive rate	Use when false positives are more expensive than false negatives.
Precision	Use when it's very important for positive predictions to be accurate.

– The F1 Score is the harmonic mean of **precision** and **recall**.

– It is useful when we need a balance between precision and recall as it combines both into a single number.

– A high F1 score means the model performs well on both metrics. Its range is [0,1].

– More the F1 score better will be performance.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

– The ROC curve is drawn by calculating the true positive rate (TPR) and false positive rate (FPR) at every possible threshold.

– The curve helps us visualize the trade-offs between sensitivity (TPR) and specificity (1 - FPR) across various thresholds.

– Area Under Curve (AUC) quantifies the overall ability of the model to distinguish between positive and negative classes.

– AUC = 1: Perfect model.

– AUC = 0.5: Model performs no better than random guessing.

– AUC < 0.5: Model performs worse than random guessing.